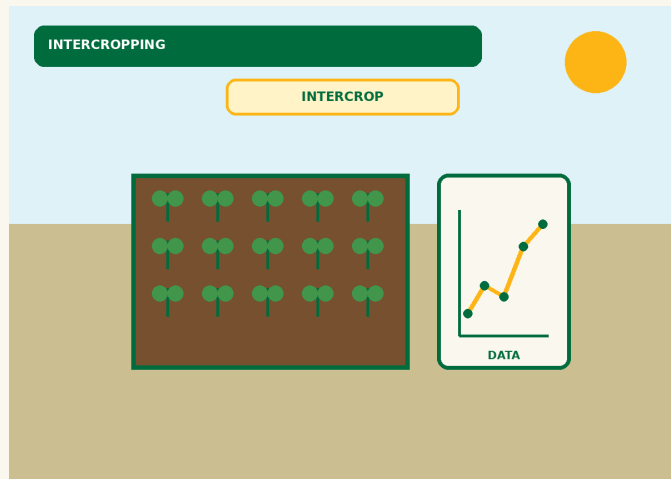




INTERCROP

INTERCROPPING

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 681 • INTERCROPPING

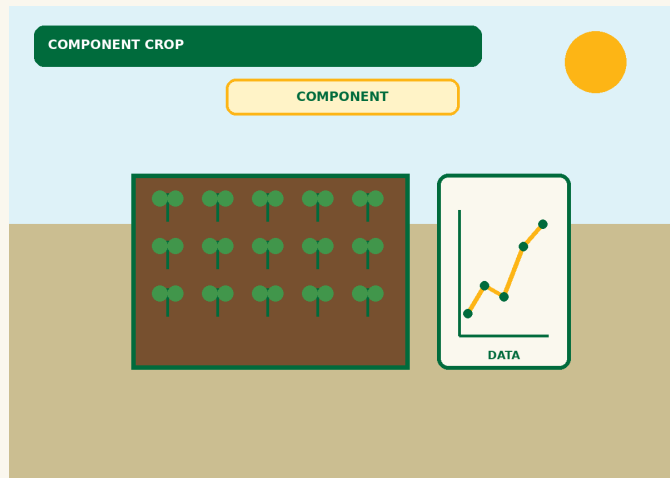
CONCEPT

Intercropping is the planned cultivation of two or more crop species in the same area for at least part of their growth cycles.

INVESTIGATION TASK

Identify the component crops and arrangement.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



COMPONENT

COMPONENT CROP

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 682 • COMPONENT CROP

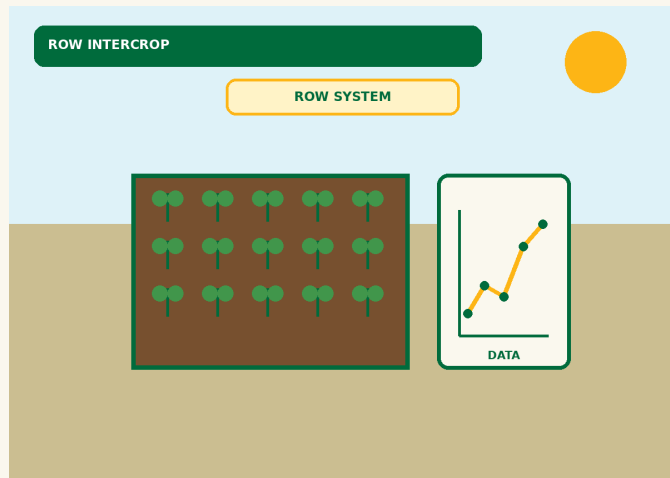
CONCEPT

A component crop is one of the species deliberately included in an intercrop system.

INVESTIGATION TASK

Describe the role of each component.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



ROW SYSTEM

ROW INTERCROP

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 683 • ROW INTERCROP

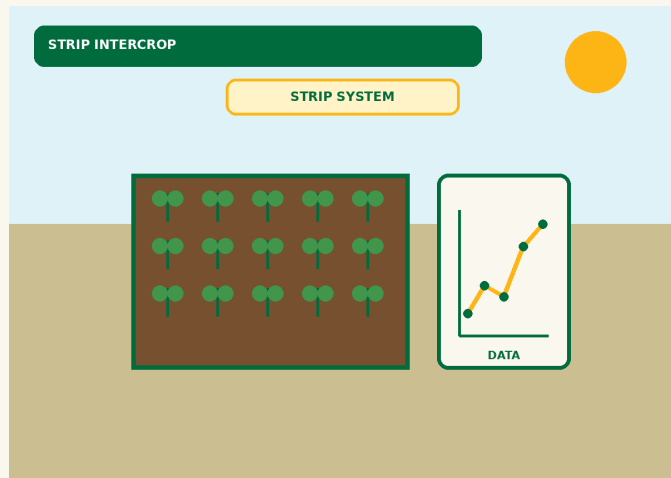
CONCEPT

Row intercropping places component crops in distinct adjacent rows.

INVESTIGATION TASK

Measure the row arrangement.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



STRIP SYSTEM

STRIP INTERCROP

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 684 • STRIP INTERCROP

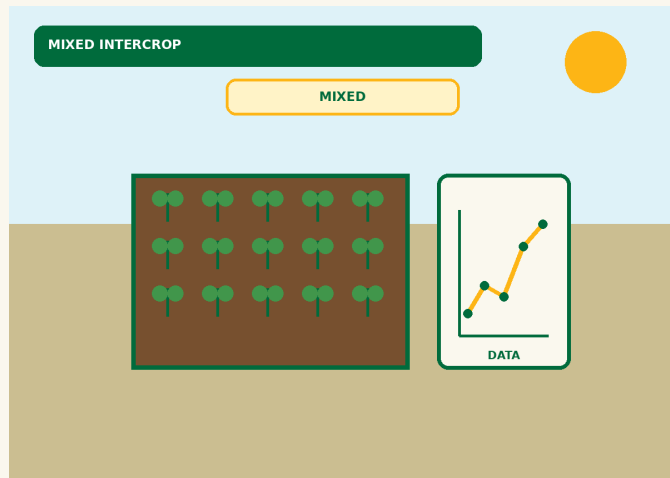
CONCEPT

Strip intercropping grows component crops in wider strips that remain close enough to interact.

INVESTIGATION TASK

Sketch a possible strip layout.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MIXED

MIXED INTERCROP

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 685 • MIXED INTERCROP

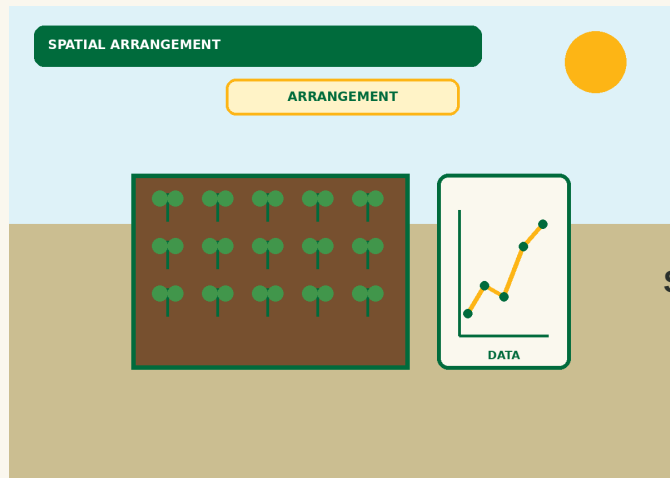
CONCEPT

Mixed intercropping combines crops without clearly separated rows.

INVESTIGATION TASK

Compare its management needs with row intercropping.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



ARRANGEMENT

SPATIAL ARRANGEMENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 686 • SPATIAL ARRANGEMENT

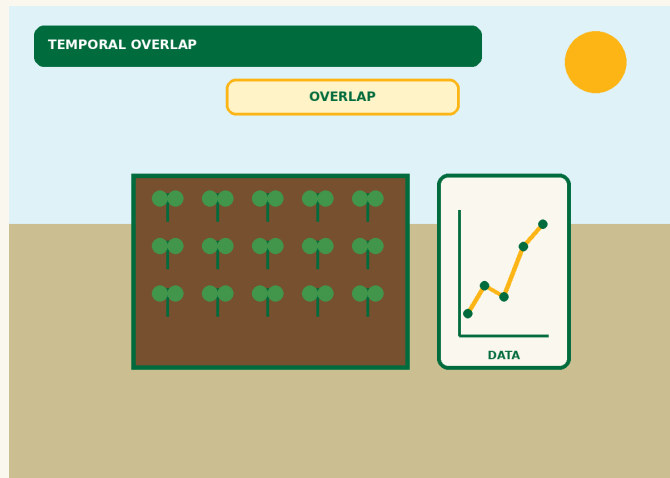
CONCEPT

Spatial arrangement determines where component crops occupy horizontal and vertical space.

INVESTIGATION TASK

Map plant positions in a sample plot.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



OVERLAP

TEMPORAL OVERLAP

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 687 • TEMPORAL OVERLAP

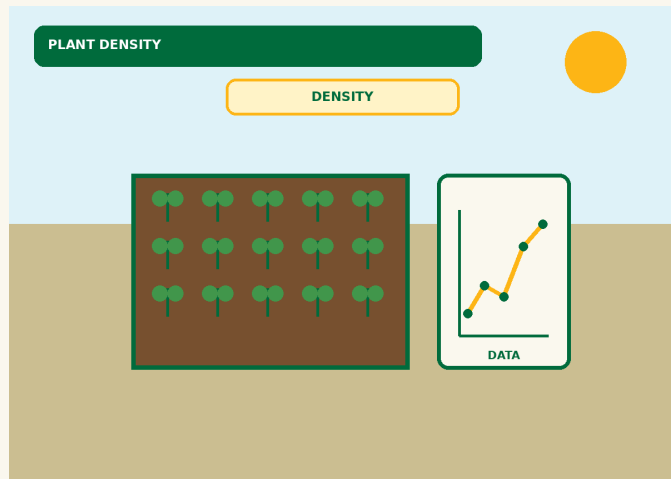
CONCEPT

Temporal overlap describes how long component crops grow together during the production cycle.

INVESTIGATION TASK

Calculate overlap from planting and harvest dates.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



DENSITY

PLANT DENSITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 688 • PLANT DENSITY

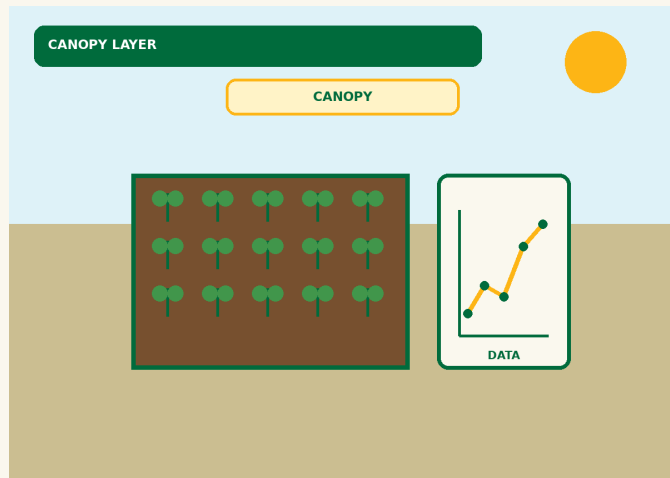
CONCEPT

Intercrop density must account for the populations of all component crops rather than one crop alone.

INVESTIGATION TASK

Calculate component plant populations.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



CANOPY

CANOPY LAYER

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 689 • CANOPY LAYER

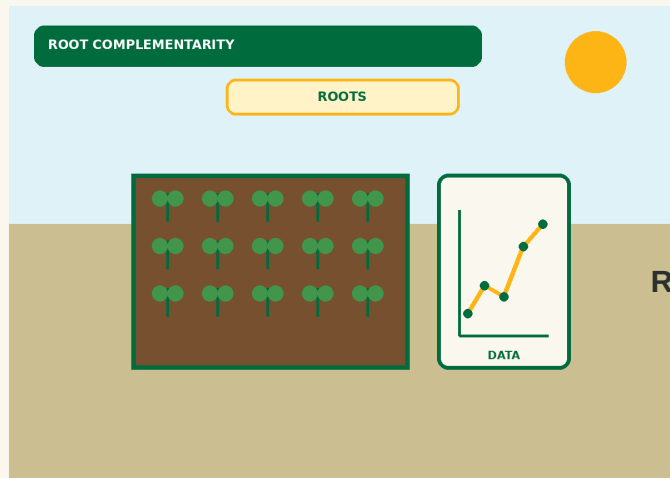
CONCEPT

Different crop heights and leaf arrangements can create vertical canopy layers.

INVESTIGATION TASK

Compare canopy positions of two crops.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



ROOTS

ROOT COMPLEMENTARITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 690 • ROOT COMPLEMENTARITY

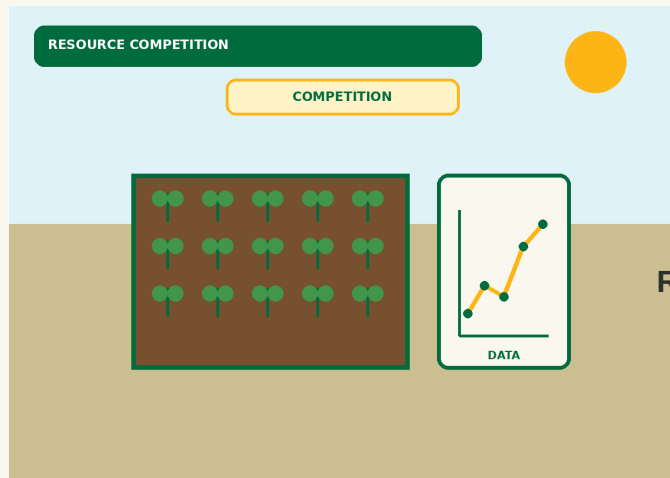
CONCEPT

Crops with different root distributions may explore soil resources differently in space or depth.

INVESTIGATION TASK

Use root diagrams to compare component crops.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



COMPETITION

RESOURCE COMPETITION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 691 • RESOURCE COMPETITION

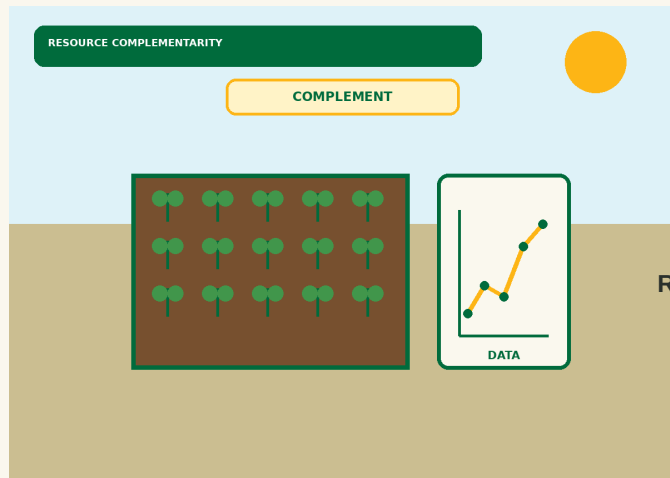
CONCEPT

Component crops may compete for light, water, nutrients, and space when demands overlap.

INVESTIGATION TASK

Identify one likely competition pathway.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



COMPLEMENT

RESOURCE COMPLEMENTARITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 692 • RESOURCE COMPLEMENTARITY

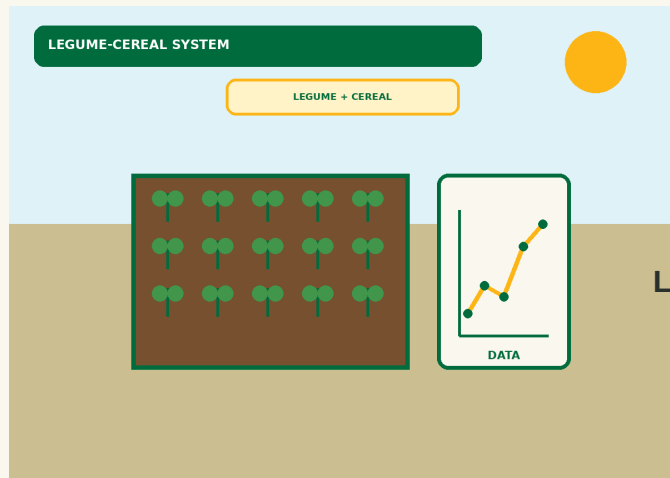
CONCEPT

Complementarity occurs when component crops use resources differently enough to reduce direct competition or improve combined capture.

INVESTIGATION TASK

Describe one plausible complementary interaction.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



LEGUME + CEREAL

LEGUME-CEREAL SYSTEM

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 693 • LEGUME-CEREAL SYSTEM

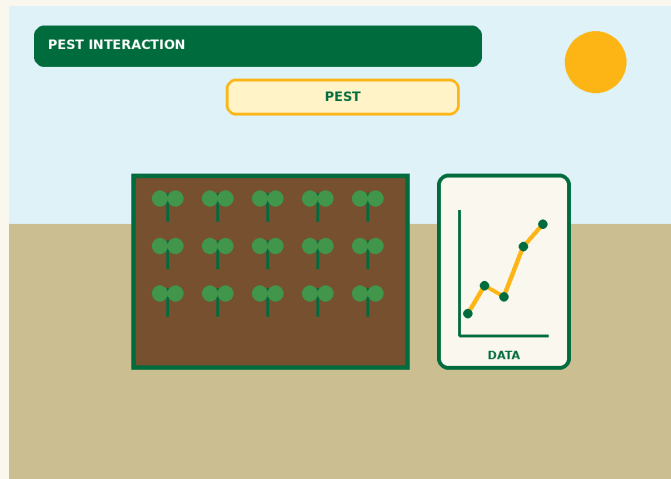
CONCEPT

Legume-cereal intercrops combine crops with contrasting growth and nutrient-acquisition characteristics.

INVESTIGATION TASK

Compare the roles of both crops.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



PEST

PEST INTERACTION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 694 • PEST INTERACTION

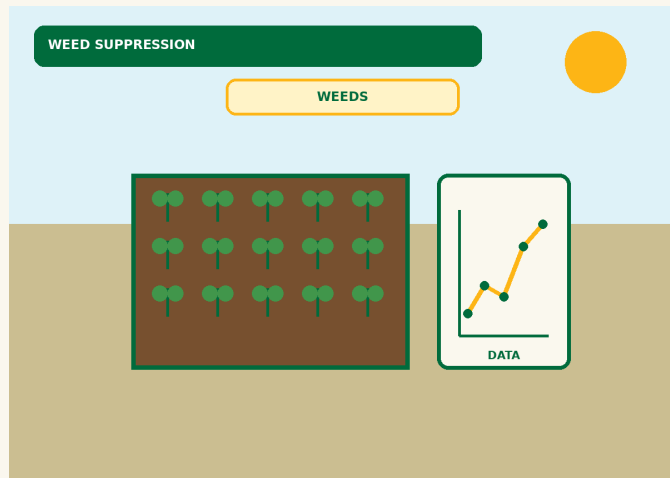
CONCEPT

Crop diversity can alter pest movement, host finding, habitat, and natural-enemy activity, but effects depend on context.

INVESTIGATION TASK

Record pest observations by component crop.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



WEEDS

WEED SUPPRESSION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 695 • WEED SUPPRESSION

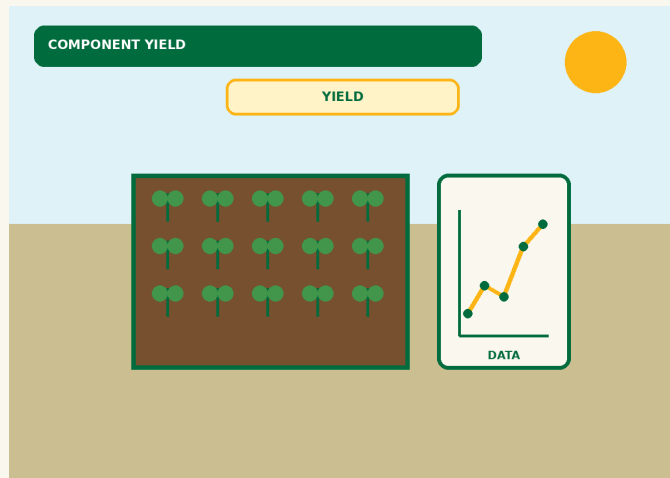
CONCEPT

A dense or layered intercrop canopy may reduce space and light available to some weeds.

INVESTIGATION TASK

Compare weed cover in two plots.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



YIELD

COMPONENT YIELD

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 696 • COMPONENT YIELD

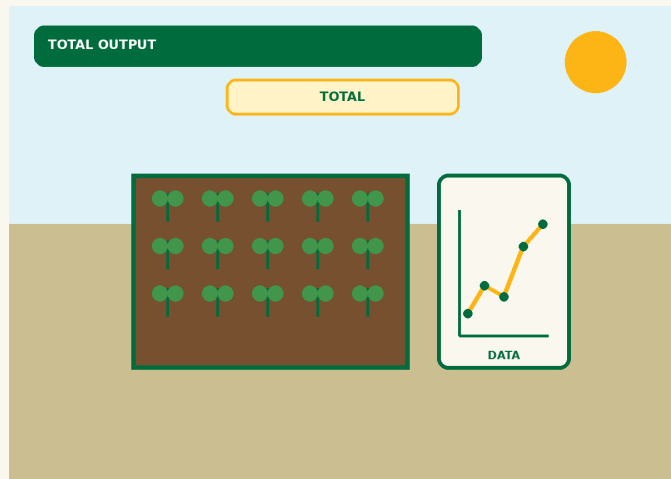
CONCEPT

Each component crop should be harvested and measured separately before combined system performance is evaluated.

INVESTIGATION TASK

Record yields by component.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



TOTAL

TOTAL OUTPUT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 697 • TOTAL OUTPUT

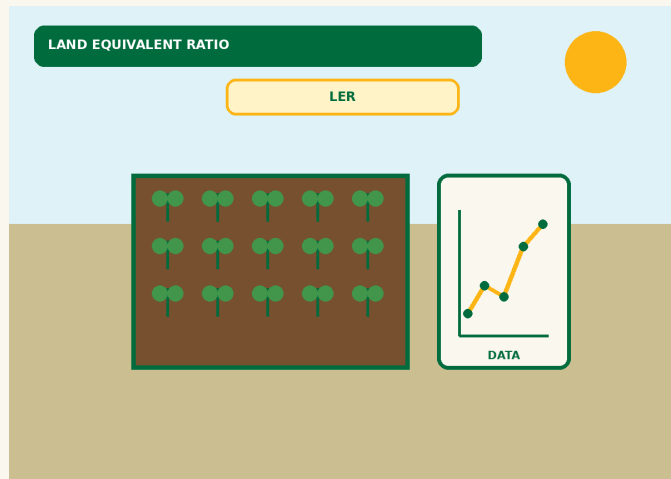
CONCEPT

Total intercrop output combines the useful products from all component crops.

INVESTIGATION TASK

Calculate total harvested mass for a sample plot.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



LER

LAND EQUIVALENT RATIO

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 698 • LAND EQUIVALENT RATIO

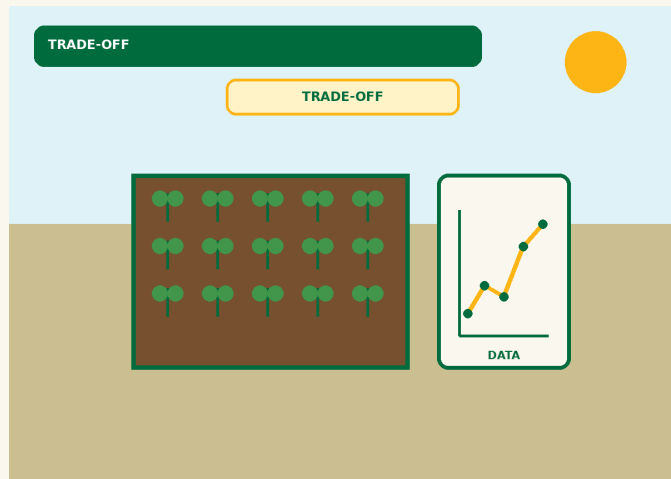
CONCEPT

Land Equivalent Ratio compares the relative land area needed under sole crops to match intercrop yields; values above 1 indicate a land-use advantage under the compared conditions.

INVESTIGATION TASK

Calculate a simple teacher-provided LER example.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



TRADE-OFF

TRADE-OFF

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 699 • TRADE-OFF

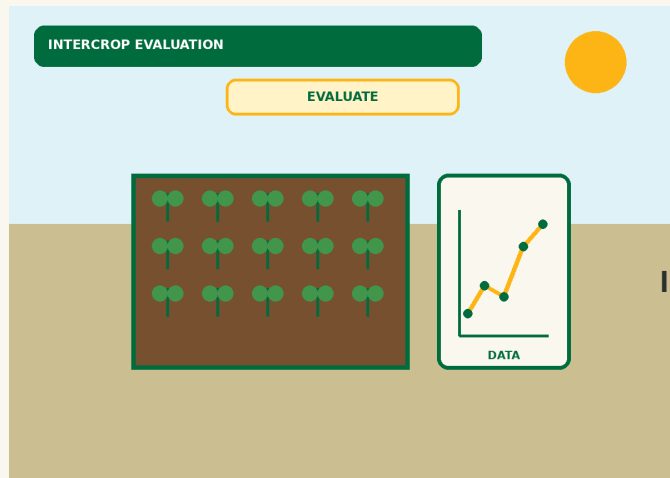
CONCEPT

A trade-off occurs when improvement in one component or objective is associated with a reduction in another.

INVESTIGATION TASK

Identify one possible intercrop trade-off.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



EVALUATE

INTERCROP EVALUATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 700 • INTERCROP EVALUATION

CONCEPT

Intercrop evaluation considers arrangement, component yields, competition, complementarity, labor, risk, and land-use performance.

INVESTIGATION TASK

Recommend whether to retain or modify the system.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE